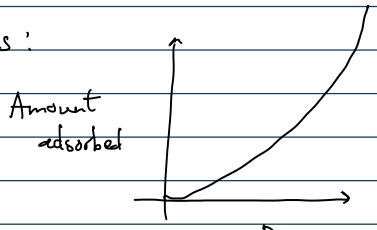
Lecture 15

Thursday, 26 October 2023 10:35 AM

Adsorption Isotherms:



Adsorbate-adsorbent interactions (favorable) ✓ } Presence of other adsorbates increases coverage

N_2 → quadrupole moment

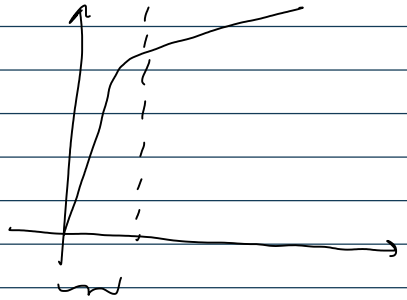
Hysteresis: material remembers memory of what the adsorption was.

MOFs: BET equation is not always valid for them

↓

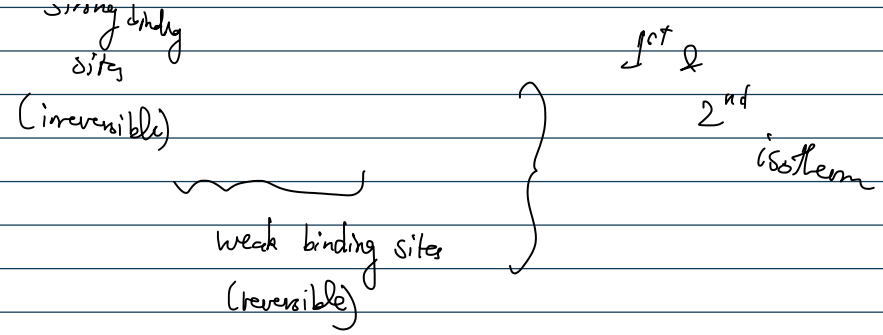
confusion b/w mesopores & micropores by the equation.

Chemisorption: quantifying surface atoms/ active sites (not just how many pores we have)



Strong binding sites

1st &



Complications: + Changes in surface due to adsorption

Support vs. Metal:

→ Ads on support vs. on weak site on support

→ Exp.: Henry's law from CO on support
(slope)
↳ Then compare slope w/ ads. slopes to say from support or metal.

Adsorbate-induced reconstruction → can draw the metal out!